SI Units and Unit Conversions

Base units, prefixes and dimensional analysis

Name _____ Class _____ Date _____

PREFIXES AND EQUALITIES

$k = 10^{3} \cdot c = 10^{-2} \cdot m = 10^{-3} \cdot \mu = 10^{-6} \cdot n = 10^{-9} \cdot G = 10^{9}$

$1\ mL = 1\ cm^{3} \cdot 1\ L = 1000\ mL = 1\ dm^{3} \cdot 1\ h = 3600\ s$ Density of water = 1.00 g/mL

Write every conversion factor with units and show which units cancel.

A. Units and prefixes

1. Give the SI base unit (name and symbol) for (a) mass (b) length (c) time (d) temperature (e) amount of substance.

2. Write each as a power of ten of the base unit: (a) 1 µg (b) 1 GHz (c) 1 nm

B. One-step conversions

3. Convert: (a) 2.5 km to m (b) 350 mL to L (c) 45 mg to g (d) 0.0045 kg to mg (e) 500. nm to m

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C. Areas and volumes

4. Convert: (a) 5.0 m^2 to cm^2 (b) 250 cm^3 to m^3 (c) 1.0 m^3 to L (d) 0.250 L to cm^3

D. Multi-step conversions

5. Convert 1.5 h to seconds.

6. A cyclist rides at 36 km/h. Express this speed in m/s.

7. The density of iron is 7.86 g/cm^3. Express it in kg/m^3.

8. What is the mass of 15.0 mL of mercury (density 13.6 g/mL)?

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E. Challenge

9. A swimming pool measures 25.0 m × 10.0 m × 2.00 m. Calculate (a) its volume in m^3 and in L and (b) the mass of water it holds, in kg.

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